

Your average CPU air cooler

Your average CPU cooler has very few components. There's the baseplate which sits atop the CPU and draws heat from the CPU. Then there is the heatsink which dissipates all the heat. If you're using a custom cooler then you'll have heatpipes or a vapour chamber to accelerate the heat transfer. The heatpipes have fins attached to them and the fins, in turn, have a fan attached to them to drive air and further accelerate the cooling. This is just the superficial way of looking at air coolers. Let's go a bit into details.

Base plate

Like all coolers, the base plate is what makes contact with the CPU. It's made of different materials with copper and aluminium being the most common. Copper can sometimes be tin-plated to reduce corrosion and ensure longevity. The same goes for the fins as well. There are a lot of different theories regarding the effect of the surface finish of the base plate with the cooling efficacy or, should we say, heat transfer efficacy. Some hold the opinion that a glossy mirror finish is better since it mates better with the CPU surface while some hold the opinion that a matte / brushed finish is better since it offers more surface area. While the logic behind both approaches are sound, the simple fact is that the difference between the two approaches is negligible.

Having a flat surface that mates well with the CPU and the amount of material between the CPU IHS (Integrated Heat Sink) and the cooler base plate are what matter the most.

The reason for the former is plain and straightforward since conduction happens better when the surfaces are mated well. As for the latter, the same logic applies. If you have direct contact with the material that conducts the most heat, then you can cool faster. This is why a lot of coolers with heatpipes tend to ensure that the heatpipes are in direct contact with the CPU surface. However, doing so also comes at a disadvantage. Since the heatpipes are cylindrical and the CPU surface is flat, you end up with tiny grooves on the cooler base plate which doesn't remain flush with the CPU surface. As a result, the cooling efficacy



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